SkyGuard AI - Frontend Requirements

For the hackathon dashboard. Keep the UI focused on live monitoring, anomaly detection, explanation, and sensor health.

Core dashboard sections

Section	What to show
Live sensor readings	Temperature (°C), Relative Humidity (%), Pressure (hPa), and last-updated timestamp.
Overall AWS status	Healthy / Warning / Critical status for the whole station.
Real-time anomaly alert	Sensor, anomaly type, current reading, severity, confidence, and alert timestamp.
Live graphs	Temperature vs time, Humidity vs time, Pressure vs time. Highlight anomalous points.
Sensor health cards	Separate health status for Temperature, Humidity, and Pressure sensors.
Anomaly explanation	Simple reason for the alert, e.g. sudden temperature jump or abnormal combination of readings.
Root-cause classification	Spike, Frozen Sensor, Drift, Communication Error, or Multivariate Inconsistency.
Anomaly history	Table of past alerts with time, sensor, reading, anomaly type, and severity.
Anomaly/confidence score	A clear score or gauge showing how suspicious the current reading is.
Maintenance recommendation	Suggested action such as inspect, recalibrate, or check sensor connection.
Optional corrected value	Show estimated corrected reading beside the faulty reading if this feature is implemented.

Priority for the first working version

Must have: Live readings, live graphs, anomaly alert, anomaly type, severity/confidence, sensor health, anomaly explanation, and anomaly history.

Suggested alert format

ANOMALY DETECTED
Sensor: Temperature
Type: Sudden Spike
Reading: 52.4 °C
Severity: High
Confidence: 94%
Reason: Temperature changed too quickly compared with recent 24-hour behaviour.

Data the frontend should expect from backend

timestamp, temperature, humidity, pressure, overall_status, anomaly_status, anomaly_score, severity, anomaly_type, affected_sensor, explanation, sensor_health, and optional corrected_value.

Design goal: A judge should be able to look at the dashboard for 5 seconds and understand: current weather readings, whether the station is healthy, what went wrong, and why the AI flagged it.